

```
In [1]: import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
from sklearn.model_selection import train_test_split
from sklearn.neighbors import KNeighborsClassifier
from sklearn.metrics import classification_report, accuracy_score
```

```
In [2]: data = sns.load_dataset('iris')
data.head()
```

```
Out[2]:
```

	sepal_length	sepal_width	petal_length	petal_width	species
0	5.1	3.5	1.4	0.2	setosa
1	4.9	3.0	1.4	0.2	setosa
2	4.7	3.2	1.3	0.2	setosa
3	4.6	3.1	1.5	0.2	setosa
4	5.0	3.6	1.4	0.2	setosa

```
In [4]: type(data)
```

```
Out[4]: pandas.DataFrame
```

```
In [5]: data.shape
```

```
Out[5]: (150, 5)
```

```
In [6]: data.species.value_counts()
```

```
Out[6]: species
setosa      50
versicolor  50
virginica   50
Name: count, dtype: int64
```

```
In [7]: data.isna().sum()
```

```
Out[7]: sepal_length    0
sepal_width          0
petal_length         0
petal_width          0
species              0
dtype: int64
```

```
In [34]: data.iloc[100:150]
```

Out[34]:

	sepal_length	sepal_width	petal_length	petal_width	species
100	6.3	3.3	6.0	2.5	virginica
101	5.8	2.7	5.1	1.9	virginica
102	7.1	3.0	5.9	2.1	virginica
103	6.3	2.9	5.6	1.8	virginica
104	6.5	3.0	5.8	2.2	virginica
105	7.6	3.0	6.6	2.1	virginica
106	4.9	2.5	4.5	1.7	virginica
107	7.3	2.9	6.3	1.8	virginica
108	6.7	2.5	5.8	1.8	virginica
109	7.2	3.6	6.1	2.5	virginica
110	6.5	3.2	5.1	2.0	virginica
111	6.4	2.7	5.3	1.9	virginica
112	6.8	3.0	5.5	2.1	virginica
113	5.7	2.5	5.0	2.0	virginica
114	5.8	2.8	5.1	2.4	virginica
115	6.4	3.2	5.3	2.3	virginica
116	6.5	3.0	5.5	1.8	virginica
117	7.7	3.8	6.7	2.2	virginica
118	7.7	2.6	6.9	2.3	virginica
119	6.0	2.2	5.0	1.5	virginica
120	6.9	3.2	5.7	2.3	virginica
121	5.6	2.8	4.9	2.0	virginica
122	7.7	2.8	6.7	2.0	virginica
123	6.3	2.7	4.9	1.8	virginica
124	6.7	3.3	5.7	2.1	virginica
125	7.2	3.2	6.0	1.8	virginica
126	6.2	2.8	4.8	1.8	virginica
127	6.1	3.0	4.9	1.8	virginica
128	6.4	2.8	5.6	2.1	virginica
129	7.2	3.0	5.8	1.6	virginica

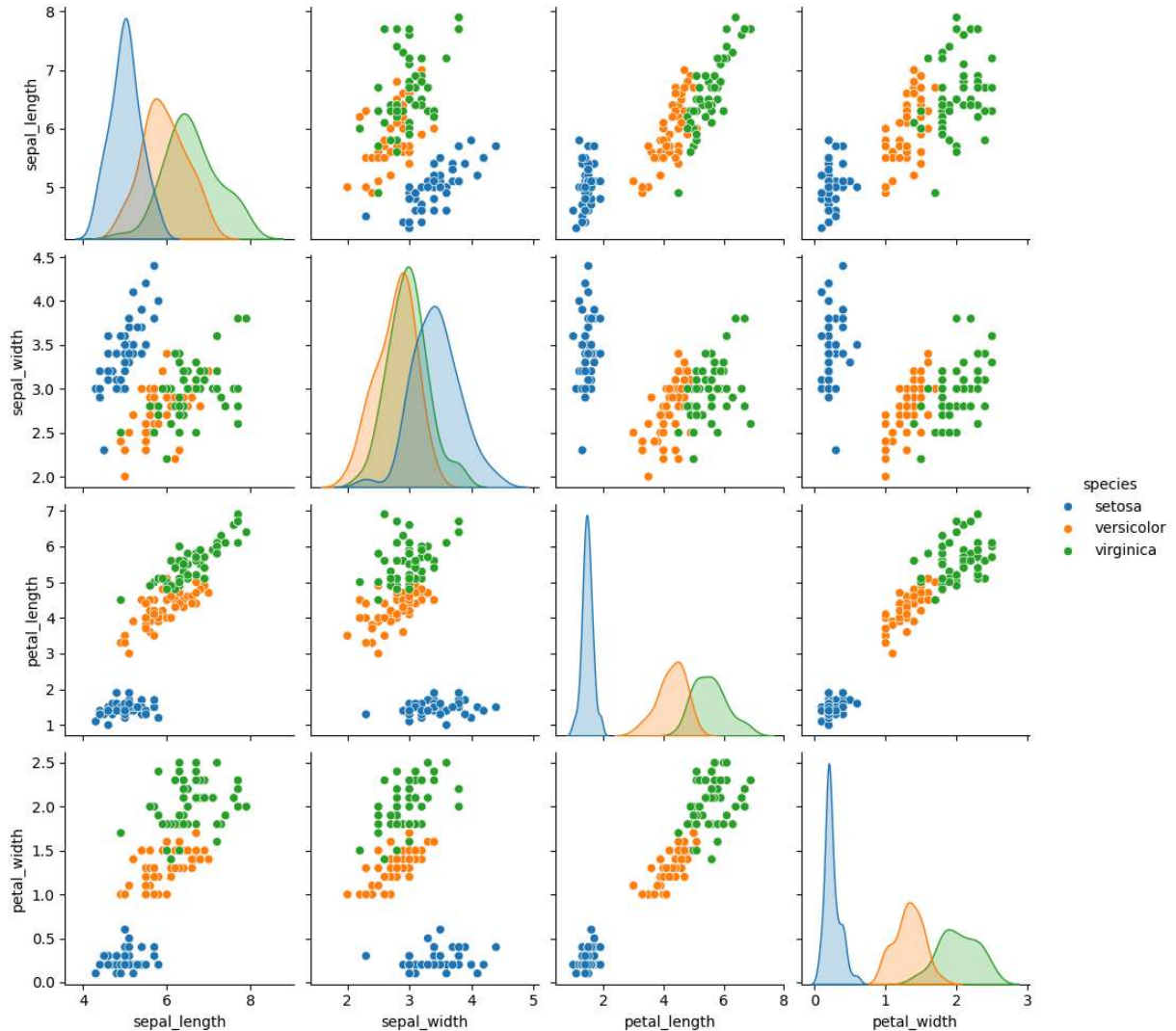
	sepal_length	sepal_width	petal_length	petal_width	species
130	7.4	2.8	6.1	1.9	virginica
131	7.9	3.8	6.4	2.0	virginica
132	6.4	2.8	5.6	2.2	virginica
133	6.3	2.8	5.1	1.5	virginica
134	6.1	2.6	5.6	1.4	virginica
135	7.7	3.0	6.1	2.3	virginica
136	6.3	3.4	5.6	2.4	virginica
137	6.4	3.1	5.5	1.8	virginica
138	6.0	3.0	4.8	1.8	virginica
139	6.9	3.1	5.4	2.1	virginica
140	6.7	3.1	5.6	2.4	virginica
141	6.9	3.1	5.1	2.3	virginica
142	5.8	2.7	5.1	1.9	virginica
143	6.8	3.2	5.9	2.3	virginica
144	6.7	3.3	5.7	2.5	virginica
145	6.7	3.0	5.2	2.3	virginica
146	6.3	2.5	5.0	1.9	virginica
147	6.5	3.0	5.2	2.0	virginica
148	6.2	3.4	5.4	2.3	virginica
149	5.9	3.0	5.1	1.8	virginica

```
In [10]: data.describe()
```

Out[10]:

	sepal_length	sepal_width	petal_length	petal_width
count	150.000000	150.000000	150.000000	150.000000
mean	5.843333	3.057333	3.758000	1.199333
std	0.828066	0.435866	1.765298	0.762238
min	4.300000	2.000000	1.000000	0.100000
25%	5.100000	2.800000	1.600000	0.300000
50%	5.800000	3.000000	4.350000	1.300000
75%	6.400000	3.300000	5.100000	1.800000
max	7.900000	4.400000	6.900000	2.500000

```
In [12]: sns.pairplot(data, hue='species')
plt.show()
```



```
In [13]: data.head(3)
```

Out[13]:

	sepal_length	sepal_width	petal_length	petal_width	species
0	5.1	3.5	1.4	0.2	setosa
1	4.9	3.0	1.4	0.2	setosa
2	4.7	3.2	1.3	0.2	setosa

In [14]: `X = data.drop('species', axis='columns')`
`X.head(2)`

Out[14]:

	sepal_length	sepal_width	petal_length	petal_width
0	5.1	3.5	1.4	0.2
1	4.9	3.0	1.4	0.2

In [15]: `y = data.species`
`y.head(2)`

Out[15]:

```
0    setosa
1    setosa
Name: species, dtype: str
```

In [16]: `X_train,X_test,y_train,y_test = train_test_split(X,y,test_size=0.2, random_state=42)`

In [17]: `knn = KNeighborsClassifier(n_neighbors = 3)`
`knn.fit(X_train,y_train)`

Out[17]:

▼ KNeighborsClassifier ⓘ ?

► Parameters

► Fitted attributes

In [18]: `y_pred = knn.predict(X_test)`
`y_pred`

Out[18]:

```
array(['versicolor', 'setosa', 'virginica', 'versicolor', 'versicolor',
       'setosa', 'versicolor', 'virginica', 'versicolor', 'versicolor',
       'virginica', 'setosa', 'setosa', 'setosa', 'setosa', 'versicolor',
       'virginica', 'versicolor', 'versicolor', 'virginica', 'setosa',
       'virginica', 'setosa', 'virginica', 'virginica', 'virginica',
       'virginica', 'virginica', 'setosa', 'setosa'], dtype=object)
```

In [19]: `print("Accuracy: ", accuracy_score(y_test,y_pred))`

Accuracy: 1.0

In [25]: `print(classification_report(y_test,y_pred))`

	precision	recall	f1-score	support
setosa	1.00	1.00	1.00	10
versicolor	1.00	1.00	1.00	9
virginica	1.00	1.00	1.00	11
accuracy			1.00	30
macro avg	1.00	1.00	1.00	30
weighted avg	1.00	1.00	1.00	30

```
In [26]: # Lets test prediction
data.head(2)
```

```
Out[26]:
```

	sepal_length	sepal_width	petal_length	petal_width	species
0	5.1	3.5	1.4	0.2	setosa
1	4.9	3.0	1.4	0.2	setosa

```
In [29]: new_data = pd.DataFrame({'sepal_length':[5.1], 'sepal_width':3.5, 'petal_length':1.4,
new_data
```

```
Out[29]:
```

	sepal_length	sepal_width	petal_length	petal_width
0	5.1	3.5	1.4	0.2

```
In [31]: prediction = knn.predict(new_data)
prediction[0]
```

```
Out[31]: 'setosa'
```

```
In [33]: # 5.7  2.8  4.1  1.3  versicolor
new_data = pd.DataFrame({'sepal_length':[5.7], 'sepal_width':2.8, 'petal_length':4.1,
knn.predict(new_data)[0]
```

```
Out[33]: 'versicolor'
```

```
In [35]: # 6.5  3.0  5.2  2.0  virginica
new_data = pd.DataFrame({'sepal_length':[6.5], 'sepal_width':3.0, 'petal_length':5.2,
knn.predict(new_data)[0]
```

```
Out[35]: 'virginica'
```

```
In [36]: import joblib
```

```
In [37]: joblib.dump(knn, f"iris_flower_class_model.pkl")
```

```
Out[37]: ['iris_flower_class_model.pkl']
```

```
In [ ]:
```